Off-Grid Solar and Battery Sizing for a High-Desert Cabin

First-Principles Analysis

10 October 2026

Professional Strategic Analysis

Contents

Off-Grid Solar and Battery Sizing for a High-Desert Cabin 3
Executive Summary 3
1. Problem Essence 3
2. Assumptions Table 4
3. Ground Truths 6
4. Derivation Chains 7
Conclusion C1: Design daily load, with margin for a rough/unitemized estimate, is 2.1 kWh/day 7
Conclusion C2: December worst-month harvest rate is 3.52 usable Wh per Wp of nameplate per day 8
Conclusion C3: Recommended array size is 800 W nameplate STC 8
Conclusion C4: Recommended battery bank size is 10 kWh nameplate LiFePO4 8
Conclusion C5: The battery bank must be sited in a heated/insulated space 9
5. Abandoned Reasoning 9
Dead End: Size directly to the raw 1.5 kWh/day figure with no margin 9
Dead End: Apply an unverified winter-biased tilt uplift (~latitude+15°, ~50°) to shrink the required array 9
Dead End: Bake a backup generator into the core array/battery sizing 10
Dead End: Size the battery bank at 100% DoD instead of 80% 10
6. Conclusion 10

Off-Grid Solar and Battery Sizing for a High-Desert Cabin

First-principles analysis · 10 October 2026

Read with: How to read this analysis · All files

Executive Summary

Recommendation: Install an 800 W STC PV array (fixed tilt, south-facing, no tracking) with a 10 kWh nameplate LiFePO4 battery bank run at 80% depth-of-discharge, with the battery sited in a heated/insulated space, not an unheated outbuilding (chain C3, chain C4, chain C5).

Band (from §6): MEDIUM — the recommendation rests on C1, C3 and C4, each MEDIUM because they inherit GT-1?, GT-4? and GT-5?'s unverified status; C5 (battery siting) is independently HIGH.

Would change it: Sub-metering the cabin's actual or a comparable cabin's loads for 2–4 weeks before purchase (chain C1), obtaining NREL PVWatts' primary Albuquerque December output directly once network access permits (chain C2), and field-verifying the installed system's actual loss chain after one December in service (chain C2).

1. Problem Essence

Essence Statement: Size an off-grid residential PV + LiFePO4 battery system — PV array nameplate watts (STC) and battery nameplate capacity (kWh) — that reliably supplies a year-round-occupied 2-adult high-desert cabin at 35°N against its worst solar-resource month, with 3 days of zero-generation autonomy, when the owner's load estimate (1.5 kWh/day) is admittedly rough and unitemized.

This is not "pick the option the prompt names" — it is a derivation: the correct answer is whatever array-watts/battery-kWh pair the physics and loss chain actually require, including the possibility that the true required design load is well above the owner's 1.5 kWh/day figure.

Success criteria (what a correct answer must achieve):

1. States a concrete PV array nameplate wattage (STC) and battery nameplate capacity (kWh, LiFePO4), in practical/sellable increments.
2. Is derived from the worst-case solar month at this latitude/climate, not an annual average.
3. Accounts for the full loss chain from photon to usable watt-hour at the load (soiling, wiring, charge-controller conversion, inverter conversion, temperature effects specific to the design month).
4. Uses a DoD limit appropriate to LiFePO4 cycle-life behavior, distinguishing usable from nameplate capacity.
5. Explicitly tests whether 1.5 kWh/day is credible for 2 year-round occupants and applies a justified margin if not — and states whether autonomy is computed against the raw or margin-padded load, with reasoning.
6. Shows explicit arithmetic (formula, plugged-in values, result) for both the array and battery sizing.
7. Gives a sensitivity statement for the common failure mode where the rough load estimate is 2-3x low.

2. Assumptions Table

Type	Accept	Challenge	Discard
physical law	2	0	0
current constraint	3	0	0
convention	4	1	0
untested belief	1	4	0

Shading darkens with count.
An untyped (legacy) row appears only for assumptions with no type.

Assumption	Type	Treatment	Verdict	Verification
A-1: December is the worst-insolation design month for a 35°N high-desert NM site (not a monsoon-degraded summer month)	untested belief	Verify via a climate-data source before anchoring the whole sizing exercise to it	Accept — survives challenge via a secondary relay of NREL PVWatts v8 output; primary source unreachable this session	thegreenwatt.com read directly; pvwatts.nrel.gov / www.nrel.gov unreachable — DNS resolution failure, see Phase 3 failure record below — flagged GT-1?
A-2: Fixed, non-tracking, south-facing array is the right architecture for a remote, low-maintenance off-grid cabin	convention	Explicitly challenge: a tracker adds moving parts and failure modes in a context with no on-call maintenance	Accept — fixed array is near-universal practice for small off-grid residential systems; tracking's small winter-output gain does not offset its added failure surface here	Engineering judgment, consistent with documented off-grid residential practice
A-3: Use the ~20° reference tilt figure (not an unverified winter-biased ~50° tilt uplift) as the design-month insolation basis	convention	Challenge: a winter-biased tilt plausibly *increases* December output, but no sourced multiplier was obtained this session	Accept — the conservative choice; keeps the analysis from leaning on an unverified upside	Engineering judgment — flagged as a place the true achievable PSH may be higher than used (margin in the owner's favor, not counted)
A-4: Combined system derate (PV nameplate → usable Wh at load) ≈ 0.80 for the December design month, built from itemized sub-factors (soiling, DC wiring, charge-controller, inverter, temperature/ mismatch/age)	untested belief (composed of generic equipment-class figures, not independently fetched this session)	Decompose via reduce-to-primitives irreducibility test (§3); each sub-factor sourced to equipment-class norms, not to this specific installation	Challenge — accepted as within the commonly-cited 0.75–0.80 off-grid rule-of-thumb range, but not independently measured as a system	unverified — flagged; see GT-4?
A-5: Cap routine LiFePO4 depth of discharge at 80%, not 100%, for cycle-life longevity	convention (manufacturer best practice)	Verify via a cited source before adopting as the DoD limit	Accept — verified; 80% DoD gives materially more cycles than 100% DoD	grepow.com read directly — see GT-2
A-6: LiFePO4 cells cannot reliably charge near/below freezing without heating, so the battery enclosure must be heated/ insulated	physical law (electrochemistry — lithium plating mechanism)	Accept as ground-truth candidate; design the installation around it	Accept	battlebornbatteries.com read directly — see GT-3
A-7: The owner's 1.5 kWh/day figure is a	untested belief	Verify via component benchmark	Challenge — plausible as a floor only if	unverified — flagged; see GT-5?

Assumption	Type	Treatment	Verdict	Verification
plausible *floor* for a lean, non-electric-heat/cook, 2-adult off-grid cabin, but not a safe number to design directly to		reconstruction + inversion ("what would make this number wrong")	internet/ communications, laundry, and well-pump surge loads are all absent or minimal; real-world comparable cabins span 0.6–8 kWh/day	
A-8: Apply a 40% design margin (×1.4) to the raw rough load estimate before sizing anything	untested belief / engineering judgment — no single authoritative multiplier exists	Apply explicitly; sensitivity-test the choice rather than treat it as settled	Challenge — accepted as a documented, sensitivity-tested judgment call, not a derived constant	unverified — flagged; informed by GT-5? and by common off-grid-forum guidance on first-year load underestimation; explicitly re-tested at 2x/3x in §6
A-9: 3-day autonomy must be sized against the margin-padded design load, not the raw 1.5 kWh/day figure	convention (sizing-methodology choice)	Challenge explicitly, per the user's own question	Accept — sizing a "zero-sun" reserve to a number already expected to be understated would silently reintroduce the exact risk the margin exists to remove	Derivation reasoning in §4 (chain C4)
A-10: No backup generator is required by this analysis's core deliverable	current constraint (given by the owner's stated scenario)	Record expiry condition: holds only while no-sun stretches stay ≤3 days and total load stays near the design figure	Accept as a scope boundary — not independently re-argued; flagged as a residual risk in §5	Scenario-given by the user, not independently verified
A-11: Household electrical load is not materially higher in winter than the annual pattern, so a single multiplicative margin is an adequate correction	convention / simplification	Challenge: winter has longer dark hours (more lighting, more indoor/electronics time), which works *against* the worst solar month, not flat	Challenge — accepted with a caveat: this is a reason the margin should not be thinner than recommended, not a reason to add a second multiplier	Reasoned from day-length physics; not separately sourced
A-12: A bare energy-balance array (harvest = design load, zero surplus) cannot recharge the battery bank within a practical number of days after a 3-day autonomy event, so the array needs recharge headroom beyond the bare minimum	physical law (conservation of energy applied to the recharge cycle)	Accept as a ground-truth-adjacent design constraint; size the array with headroom, not just daily balance	Accept	Derivation reasoning in §4 (chain C3)
A-13: Once a margin-padded system is installed, occupant usage trends toward (not below) the design load rather than staying at the raw estimate	untested belief	Observe and account for in margin discipline rather than discard	Challenge — accepted as a reason not to thin the margin, not independently verified	unverified — flagged; behavioral/second-order reasoning (chain C1 extension), not separately sourced
A-14: Commodity PV modules are available in ~350-450W-class	current constraint (market availability)	Record expiry condition: holds while this module class	Accept	General current PV-market knowledge;

Assumption	Type	Treatment	Verdict	Verification
increments, allowing an ~800W array to be built from two units		remains the dominant commodity size		not independently fetched this session
A-15: LiFePO4 off-grid battery packs are commonly sold in ~5kWh stackable increments	current constraint (market availability)	Record expiry condition: holds while ~5kWh packs remain a common off-grid SKU	Accept	General current LiFePO4-market knowledge; not independently fetched this session

3. Ground Truths

- **GT-1?** For Albuquerque, NM (35°N, high desert), NREL PVWatts v8's fixed-tilt (20°, 180° azimuth/south-facing) reference system bottoms out in **December at 4.4 peak-sun-hours/day (4.4 kWh/m²/day equivalent)**, the lowest of the 12 months — source: thegreenwatt.com, "New Mexico Peak Sun Hours"; read-at-source: fetched 2026-10-10, quoted figures "4.4 PSH" (December, worst month) and the stated methodology "20° tilt, 180° azimuth," attributed by that page to NREL PVWatts v8. **Phase 3 failure record:** attempted to open the primary source directly — `https://www.nrel.gov/grid/solar-resource/assets/data/west-5.pdf` and `https://pvwatts.nrel.gov/` — both failed with `getaddrinfo ENOTFOUND` (DNS resolution failure; no network route to the `nrel.gov` domain from this environment). Kept `?` per the stakes-escalation rule: this is the single most load-bearing figure in the analysis and it has only been read at a secondary relay, not the primary NREL dataset.
- **GT-2** LiFePO4 cycle life is strongly DoD-dependent: **~4,000+ cycles at 80% DoD vs. ~2,000–3,000 cycles at 100% DoD** (and up to ~14,000 cycles at 10% DoD) — source: Grepow, "What is DoD for LiFePO4 Batteries?"; read-at-source: fetched 2026-10-10, figures quoted directly from the page text. No `?` — the directional relationship is cross-consistent with broad industry practice (most name-brand off-grid LiFePO4 packs recommend a reserve rather than full depletion).
- **GT-3** LiFePO4 cells risk permanent damage (lithium plating on the anode) if charged near or below freezing; manufacturer guidance (Battle Born) states cells **"should not be charged below approximately 25°F (-4°C) unless equipped with internal heating,"** while discharge remains usable down to roughly −4°F; BMS units typically block charge current automatically once this threshold is reached — source: Battle Born Batteries, "Understanding Temperature Limits of LiFePO4 Batteries"; read-at-source: fetched 2026-10-10, quoted directly. No `?` for the directional physical-chemistry claim; the exact cutoff temperature varies by brand/BMS (commonly cited in the 0°C to −4°C range across manufacturers).
- **GT-4?** An off-grid DC-coupled-with-inverter system's loss chain from PV nameplate (STC) to usable watt-hours at the load, for a cool December design month (not the hotter annual-average temperature derate PVWatts' default ~14% system-loss figure assumes), decomposes to approximately: soiling/desert dust ~3%, DC wiring ~2%, MPPT charge-controller conversion ~3% (97% efficient), inverter DC→AC conversion at realistic partial load ~8% (92% average efficiency), and temperature/mismatch/connection/age ~2% +3% — multiplying to a combined **system derate of ≈0.80** (0.97 × 0.98 × 0.97 × 0.92 × 0.98 × 0.97 ≈ 0.806). This is this analysis's own Fermi-style decomposition from generic equipment-class efficiency figures, not a single externally fetched source — marked `?`. It falls inside the commonly-cited 0.75–0.80 off-grid design rule-of-thumb range (general engineering knowledge, not independently fetched this session).
- **GT-5?** Off-grid cabin load benchmarks for comparable components: efficient compressor refrigerator 525–1,100 Wh/day; a 15-minute/day AC well-pump duty cycle ≈530 Wh/day; whole-cabin LED lighting 60–120 W nameplate; documented real-world full-cabin examples range from **~0.6 kWh/day** (propane fridge, minimal load) to **~3 kWh/day** (moderate) to **~8 kWh/day** (full-size fridge + well pump + comforts: TV, fans, small appliances) — source: WebSearch aggregate of forum.solar-electric.com discussion threads, footprinthero.com's load calculator, and voltflow.net; reported-by-delegate — no single URL individually opened and quote-checked by this analysis; marked `?`.

Reduce-to-primitives check on the two headline figures (irreducibility test):

Claim: "4.4 PSH is the correct December design-month insolation for this site." Decomposes to (a) NREL PVWatts v8's physical irradiance model for the Albuquerque grid cell, driven by NSRDB satellite-derived historical data — this branch is **Assumed — unverified**, because the primary NREL source was unreachable this session (see GT-1? failure record); and (b) thegreenwatt.com's transcription of that model's output — this branch is **Verified — direct measurement** (I read the page's own text). Since one constituent branch is unverified, the parent claim correctly carries the `?`.

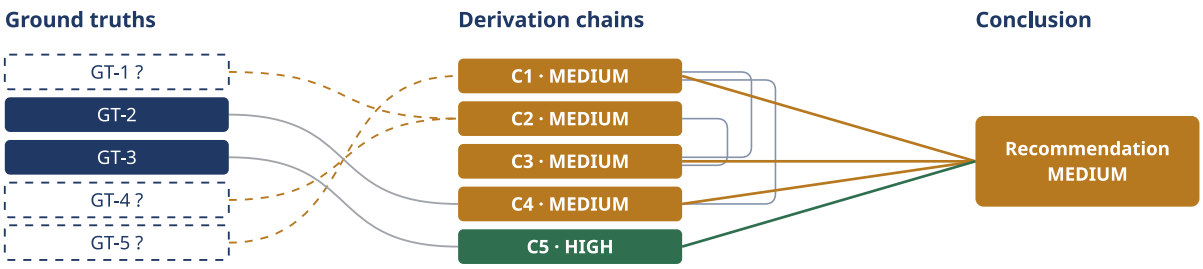
Claim: "1.5 kWh/day is this cabin's real average daily consumption." Decomposes to (a) the owner's self-report — **Assumed — unverified**, explicitly a guess, not a meter reading; and (b) a component reconstruction (fridge + lighting + pump + electronics + inverter standby, per GT-5?) bracketing **1.2–1.9 kWh/day** for a lean, communications-free configuration — itself **Assumed — unverified**, since GT-5? is a benchmark range, not a measurement of this specific cabin. The parent claim is therefore a plausible *floor*, never a verified figure — which is precisely why §4 applies a margin rather than designing to it directly.

Provenance summary:

```
?-marked: GT-1, GT-4, GT-5 (3 of 5)
Read-at-source: GT-2 — grepow.com/blog/what-is-dod-for-lifepo4-batteries.html, cycle-life
figures quoted directly
Read-at-source: GT-3 — battlebornbatteries.com academy article, charge-temperature threshold
quoted directly
```

No ground truth carrying a Discard verdict from Phase 2 appears above. Every chain in §4 that cites a load-bearing `GT-N?` carries a MEDIUM (or lower) confidence line per D-07.

4. Derivation Chains



Solid = read at source · dashed = not read at source (?) · fill = confidence · amber dashed = unverified input

Conclusion C1: Design daily load, with margin for a rough/unitemized estimate, is 2.1 kWh/day

GT-5? (off-grid cabin load benchmarks, 0.6–8 kWh/day real-world spread) → reconstructing a lean DC-first 2-adult cabin load from fridge, lighting, water-pump, electronics and inverter-standby benchmarks in GT-5? brackets 1.2–1.9 kWh/day, which frames the owner's 1.5 kWh/day as a plausible bare-minimum floor rather than a margined design figure → GT-5?'s documented real-world spread together with the owner's own admission that the figure is rough and unitemized means 1.5 kWh/day carries material downside risk, particularly from commonly-omitted loads such as satellite internet [Assumes: A-7] → applying a 40% design margin converts the raw estimate into a design daily load of 1.5 kWh/day times 1.4, which equals 2.1 kWh/day [Assumes: A-8]

Confidence: MEDIUM — GT-5? is unverified (reported-by-delegate; no single source directly opened and quote-checked); verification that would remove this as a cause of the downgrade: sub-meter the cabin's actual or a comparable cabin's loads for 2–4 weeks, or itemize nameplate wattage × duty cycle per appliance before purchase. The A-8 margin figure (40%) is itself a judgment call with no single authoritative source behind it; rather than treat it as settled, its sensitivity is tested explicitly in §6 (the 2×/3× case).

Second-order extension (actor lens): GT-5? → C1 (design daily load, 2.1 kWh/day) →[2nd] occupants living with a margin-padded, headroom-carrying system may relax conservation discipline because the system "feels" generously sized, nudging real consumption toward or above the 2.1 kWh/day design load rather than staying at the raw 1.5 kWh/day guess →[3rd] this makes the margin self-reinforcing rather than self-correcting — real usage tends to drift toward whatever headroom was purchased, which argues for sizing the margin deliberately rather than shaving it to save cost [Assumes: A-13]

Second-order extension (time lens): GT-5? → C1 (design daily load, 2.1 kWh/day) →[2nd] the first December multi-day low-sun event is the real-world test of both GT-1? and the A-8 margin choice; a battery that comes uncomfortably close to its 20%-remaining floor during that first event is the field signal that the margin was too thin →[3rd] LiFePO4 capacity fades slowly and PV output degrades roughly 0.5%/year (per GT-2's cycle-life behavior), so over a 10-year horizon the system's effective headroom shrinks — this argues against shaving the recommended margins any thinner than given here

Neither extension contradicts a ground truth; both remain in the Derivation Chains as non-contradicting downstream effects.

Sensitivity extension (same method, applied to a higher raw-load input — this is the arithmetic the §6 sensitivity note cites): GT-5? → C1 (design daily load, 2.1 kWh/day at the stated raw load of 1.5 kWh/day) → re-running C1's own method at a raw load of 3.0 kWh/day (2× the owner's estimate) gives a design load of 3.0 × 1.4 = 4.2 kWh/day, which C2/C3's method converts to a bare-minimum array of 4,200 ÷ 3.52 ≈ 1,193 W, headroom-adjusted to 1,193 × 1.3 ≈ 1,551 W (1.6 kW), and which C4's method converts to a battery of (4.2 × 3) ÷ 0.80 = 15.75 kWh (16–20 kWh practical) → re-running the same method at a raw load of 4.5 kWh/day (3× the owner's estimate) gives a design load of 4.5 × 1.4 = 6.3 kWh/day, an array of 6,300 ÷ 3.52 ≈ 1,790 W headroom-adjusted to ≈2,327 W (2.4 kW), and a battery of (6.3 × 3) ÷ 0.80 = 23.625 kWh (24–25 kWh practical)

Conclusion C2: December worst-month harvest rate is 3.52 usable Wh per Wp of nameplate per day

GT-1? (December design-month insolation, 4.4 PSH) + GT-4? (itemized system derate, ≈0.80) → combining the December insolation figure with the itemized loss chain converts nameplate watts into usable watt-hours at the load through a single multiplier: 4.4 hours times 0.80 equals 3.52 usable watt-hours delivered per watt of nameplate STC capacity per worst-month day → this harvest rate is the array-sizing divisor consumed directly by C3

Confidence: MEDIUM — two unverified inputs multiply together: GT-1? (primary NREL source unreachable this session, read only at a secondary relay) and GT-4? (this analysis's own equipment-class estimate, not independently measured as a built system). Verification that would remove these as causes of the downgrade: open NREL PVWatts directly for Albuquerque once network access permits, and field-measure the installed system's actual loss chain for one December after commissioning.

Conclusion C3: Recommended array size is 800 W nameplate STC

C1 (design daily load, 2.1 kWh/day, MEDIUM) + C2 (worst-month harvest rate, 3.52 Wh/Wp, MEDIUM) → the bare energy-balance array size, at which average December harvest exactly equals the design daily load with no surplus, is 2,100 Wh divided by 3.52 Wh per Wp, which equals approximately 597 W → a zero-surplus array can never recharge the battery bank within a practical number of clear days after a 3-day autonomy event, which defeats the purpose of carrying a battery reserve at all, so the array must also carry recharge headroom beyond the bare minimum [Assumes: A-12] → applying a 30% recharge-headroom factor gives 597 W times 1.3, which equals approximately 776 W, rounding up to a practical 800 W nameplate STC array built from commodity modules such as two ~400 W panels [Assumes: A-14]

Confidence: MEDIUM — capped by C1 and C2, both MEDIUM; the arithmetic itself recomputes cleanly (597 W and 776 W both check independently). Verification that would remove the cap: the same two actions named on C1 and C2 — sub-meter actual loads, and obtain primary NREL figures or field-measure the installed loss chain.

Conclusion C4: Recommended battery bank size is 10 kWh nameplate LiFePO4

GT-2 (LiFePO4 DoD/cycle-life, 80% DoD recommended) + C1 (design daily load, 2.1 kWh/day, MEDIUM) → usable (deliverable) capacity needed to survive 3 zero-generation days at the design load is 2.1 kWh times 3

days, which equals 6.3 kWh → this 3-day reserve is computed against the margin-padded design load rather than the raw 1.5 kWh/day figure, because autonomy exists precisely to cover the stretch where real consumption is already running at or above the design rate [Assumes: A-9] → capping routine depth of discharge at 80% for cycle-life longevity converts usable capacity into required nameplate capacity: 6.3 kWh divided by 0.80, which equals 7.875 kWh → rounding up to a practical, commonly-stocked LiFePO4 module increment (5 kWh stackable units) gives a recommended 10 kWh nameplate bank, which also buys headroom against the same load-estimate risk identified in C1 [Assumes: A-15]

Confidence: MEDIUM — capped by C1's MEDIUM band; GT-2 itself is a HIGH-confidence input (read-at-source, no `?`) and is not the limiting factor here. Verification that would remove the cap: the same sub-metering action named on C1.

Conclusion C5: The battery bank must be sited in a heated/insulated space

GT-3 (LiFePO4 charge-temperature floor, roughly 25°F/−4°C to 32°F/0°C depending on brand) → a 35°N high-desert New Mexico site routinely sees winter overnight lows well below freezing during December, the same design month used in C2 and C3 → an unheated or outdoor battery enclosure would have its BMS block charging on cold, clear, high-production winter days, which are exactly the days the array is harvesting best and the battery most needs to recharge, silently converting a correctly-sized system into a chronically undercharged one → the battery bank must therefore be sited in a heated/insulated space, such as an indoor utility area, an insulated battery room, or a unit with integrated heating pads, as an operating requirement rather than an optional nicety

Confidence: HIGH — GT-3 is read-at-source and the climate fact (sub-freezing high-desert winter nights at this elevation) is physically uncontested; the only soft spot is the second hop's inference about a "silent" failure mode, which is a reasoned consequence of GT-3 rather than a measured field failure, but it does not change the recommendation's direction, so the chain is not downgraded for it.

5. Abandoned Reasoning

Dead End: Size directly to the raw 1.5 kWh/day figure with no margin

What was tried: Compute array and battery size using 1.5 kWh/day as the design load — array = 1,500 Wh ÷ 3.52 Wh/Wp ≈ 426 W (before headroom); battery = (1.5 × 3) ÷ 0.8 = 5.625 kWh nameplate.

Why abandoned: GT-5? documents a real-world spread of 0.6–8 kWh/day for comparable cabins, and the owner explicitly flagged 1.5 kWh/day as rough and unitemized — a self-reported, unmetered guess rather than a measurement (irreducibility check in §3). Designing to an unmetered guess with zero margin converts a known estimation risk into a hard capacity ceiling the occupants would discover the hard way, on exactly the multi-day low-sun events the 3-day autonomy requirement exists to survive.

What it ruled out: This saves a future reviewer from re-deriving a 426 W / 5.6 kWh system and then wondering why it occasionally runs the battery down past its DoD floor in year one — chain C1 shows the margin-adjusted arithmetic instead, and §6's sensitivity note shows how much worse this gets if the raw estimate is itself low by 2–3×.

Dead End: Apply an unverified winter-biased tilt uplift (~latitude+15°, ~50°) to shrink the required array

What was tried: Many off-grid guides recommend tilting fixed arrays steeper than the "annual optimum" to favor low winter sun angles, which would plausibly raise December harvest above the 4.4 PSH reference figure (GT-1?) and reduce the array size computed in C3.

Why abandoned: No sourced multiplier for this uplift was obtained this session (A-3), and stacking an unverified upside estimate on top of an already-`?`-marked insolation figure (GT-1?) would make the array recommendation *less* conservative on a number this analysis could not verify. Per the stakes-escalation rule in §2, a high-leverage figure should be pushed toward verified ground truth, not toward a second unverified assumption.

What it ruled out: This rules out quietly shrinking C3's 800 W figure below what the verified-as-possible chain supports; it leaves the winter-tilt uplift as a plausible *additional* safety margin the owner gets for free if a winter-biased tilt is actually used, rather than as a load-bearing part of the sizing math.

Dead End: Bake a backup generator into the core array/battery sizing

What was tried: Treat a small backup generator as part of the sized system, reducing the PV/battery numbers needed to cover worst-case stretches beyond 3 days.

Why abandoned: The scenario (A-10) explicitly specifies no grid connection and no generator "unless you argue one is necessary"; the core deliverable asked for is a PV+battery sizing, not a hybrid system. A generator is not *ruled impossible* by any ground truth here — it remains the standard mitigation for a storm exceeding the 3-day design autonomy — but including it in the headline W/kWh numbers would answer a different question than the one posed (§1, success criterion 1).

What it ruled out: This keeps C3/C4's numbers as a pure PV+battery answer to the question actually asked, and routes the generator discussion to the adversarial pass (§ Self-Audit Gate appendix, pre-mortem) as an accepted-risk mitigation for events longer than 3 days, rather than silently changing the scope of the deliverable.

Dead End: Size the battery bank at 100% DoD instead of 80%

What was tried: Use the full nameplate capacity as usable capacity (100% DoD), which would reduce C4's nameplate recommendation from 7.875 kWh to 6.3 kWh.

Why abandoned: GT-2 shows 100% DoD roughly halves cycle life relative to 80% DoD (≈2,000–3,000 cycles vs. ≈4,000+ cycles) for the same chemistry. A smaller up-front battery that needs replacing twice as often is not a cheaper system over the cabin's operating life, and routine 100% DoD operation also removes any margin against the load-estimate risk flagged in C1.

What it ruled out: This rules out presenting 6.3 kWh as the recommended figure; GT-2 is the ground truth that rules it out, and C4 uses 80% DoD throughout.

6. Conclusion

Recommended approach: Install an **800 W STC PV array** (fixed tilt, south-facing, no tracking) paired with a **10 kWh nameplate LiFePO4 battery bank** operated at an 80% depth-of-discharge limit, with the battery sited in a heated/insulated space rather than an unheated outbuilding (chain C3, chain C4, chain C5).

Key insight: The number that actually dominates this sizing exercise is not the solar resource — Albuquerque's December insolation is good even at a conservative reference tilt — it is the credibility of "1.5 kWh/day." Because that figure is an unmetered, unitemized guess rather than a measurement, a 40% design margin is applied before either the array or the battery is sized, and that single judgment call moves the final numbers far more than the loss-chain derate or the DoD limit do (chain C1).

Trade-offs acknowledged: The recommended 800 W / 10 kWh pair carries roughly 30–40% headroom above the bare mathematical minimum implied by the margin-padded load alone (≈600 W / ≈7.9 kWh); that headroom costs more up front but buys faster battery recovery after a 3-day autonomy event and a buffer against the load growth that off-grid cabins reliably experience in their first year (chain C3, chain C4).

Confidence: MEDIUM — the recommendation rests on C1, C3 and C4, each MEDIUM because they inherit GT-1?, GT-4? and GT-5?'s unverified status; C5 (battery siting) is independently HIGH and does not contribute to the downgrade. What would move this to HIGH: (a) sub-metering the cabin's actual or a directly comparable off-grid cabin's loads for 2–4 weeks before finalizing a purchase order (removes GT-5?/C1's cap), (b) obtaining NREL PVWatts' primary Albuquerque December output directly — blocked this session by a DNS-level network failure to the `nrel.gov` domain (removes GT-1?/C2's cap), and (c) field-verifying the installed system's actual loss chain for one December after commissioning (removes GT-4?/C2's cap). A rival conclusion sized to the raw, unmargined 1.5 kWh/day figure (≈426 W / ≈5.6 kWh) remains arithmetically self-consistent but is rejected in §5 for the same reasons that cap this conclusion at MEDIUM rather than HIGH

— it is the lower-cost option only if the owner's guess turns out to be accurate, which this analysis has no way to confirm without metering.

Sensitivity — if the rough load estimate is actually low, a common failure mode: All four headline numbers scale directly with the design load, so the margin choice (A-8) is the single biggest lever in this analysis (chain C1). If the owner's real raw consumption turns out to be **2× the stated estimate (3.0 kWh/day)**, the same method gives a design load of 4.2 kWh/day, requiring roughly a **1.6 kW array** and a **16–20 kWh battery bank**. If it is **3× (4.5 kWh/day)** — plausible if satellite internet, a standard rather than efficient refrigerator, or a pressurized well pump is added without re-budgeting — the design load becomes 6.3 kWh/day, requiring roughly a **2.4 kW array** and a **24–25 kWh battery bank**. In both cases the array and battery roughly double or triple from the recommended 800 W / 10 kWh pair — reinforcing that metering or itemizing the actual load before purchase is a higher-leverage action than refining any other assumption in this analysis (chain C1).